Modelling Aviation Maintenance Costs with SARIMA, ARIMAX and RNN: A comparison of methods

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Abstract

Accurately forecasting aircraft maintenance costs is critical for airline operators managing large and aging fleets. This study models quarterly maintenance cost per air hour for Boeing and Airbus aircraft using data from the Bureau of Transportation Statistics spanning 2004 Q_1 to 2025 Q_3. Three modeling approaches of increasing complexity are compared: a seasonal ARIMA (SARIMA) baseline, an ARIMAX model incorporating exogenous predictors including fuel costs, fleet utilization, and COVID disruption indicators, and a Long Short-Term Memory Recurrent Neural Network (LSTM-RNN). All three methods capture the upward trend in maintenance costs, though quarters associated with COVID introduce significant outliers that challenge the normality assumptions of the statistical models. Forecast accuracy on a held-out test set of seven quarters is evaluated using RMSE and MAE. Results indicate that model performance is manufacturer-dependent: the LSTM-RNN achieves the lowest error for Boeing, while the ARIMAX model performs best for Airbus, highlighting the trade-off between interpretability and predictive power in time series forecasting.

Keywords: Aviation Modeling; SARIMA; ARIMAX; Recurrent Neural Networks; COVID.

1 Introduction

Forecasting aircraft maintenance costs is a vital part of the modern supply chain. In the modern world, the companies that move millions of people and tons of cargo each day rely on the ability to accurately know both how much they will need to spend to keep their fleet up and running and when these repair costs are going to occur. This data could potentially show major disruptions, such as the COVID pandemic shutting down most commercial airline travel. Using data affected by world events presented an appealing challenge.

Disruption events are all too common in supply chain modeling yet haven't appeared in most of the data used in this course throughout the semester. This seemed like an ideal chance to test what we had learned on more chaotic data. This analysis considers three types of models, each increasing in complexity, to attempt to create effective forecasts as well as handle the extreme values seen during COVID. A SARIMA model is fitted as the baseline comparison, using only the response as a time series. The ARIMAX model attempts to expand upon this SARIMA by introducing additional predictors and aims to maintain interpretability while simultaneously increasing complexity. Finally, an RNN is trained on the data with the goal of maximizing forecasting power using all available predictors though at the cost of interpretability.

Each model is run on a similar sized training split with seven quarters of data held out for testing prediction comparison. Once fitted, each model's residuals are assessed with the knowledge that COVID may introduce more bias. Models then each forecast the most recent seven quarters of data (one full year of four quarters plus the most recent three quarters available in the data) and the errors of these predictions in the form of Root Mean Squared Error (RMSE) and Mean Average Error (MAE) are compared.

2 Data

The data used in this analysis are sourced from the Bureau of Transportation Statistics website. The first is air carrier financial data reported in form P-5.2 (Bureau of Transportation Statistics 2025a). This data includes detailed quarterly aircraft operating expenses such as payroll expenses, fuel costs, operation time, and expenditures on maintenance of equipment for large certificated U.S. air carriers. This data was last updated on 20 September 2025 and contains 66,483 quarterly entries for 201 different aircraft builds from 116 carriers over 1990 to 2025. A secondary data set containing aircraft information, such as craft type, name, and manufacturer, is also included to add context to flight data. This data set comes from the Aircraft Types data table at the Bureau of Transportation Statistics (Bureau of Transportation Statistics 2025b). This analysis will focus on maintenance expenses from 2004 onward for Airbus (Airbus Industrie) and Boeing, the two largest aircraft manufacturers represented in the data.

Once collected, the financial dataset was joined with the aircraft information dataset to stratify the financial data by either Boeing or Airbus. All rows with other manufacturers were dropped as they are not important to this analysis. Two engineered predictors were created: `MAINT_PER_AIR_HR` and `UTILISATION`. `MAINT_PER_AIR_HR` is defined as the total cost of direct maintenance divided by the total air hours flown. The raw data contains total air hours flown in thousands, so `TOTAL_AIR_HOURS` was multiplied by 1,000 to create the response, total maintenance cost per air hour (in dollars). `UTILIZATION` is defined as the total air hours flown divided by the number of days assigned to be used. This gives an indicator of how heavily a craft is used. Data was aggregated as the sum across all aircraft within each quarter for each manufacturer, resulting in a dataset of $n = 87 \times 2 = 174$ representing total costs from 2004 quarter 1 to 2025 quarter 3. This data was then further split into training and test sets for each Airbus and Boeing with n = 80 training points and n = 7 test points. The test set size was chosen such that it included a full quarter's worth of data.

The total direct cost of maintenance per air hour shows a generally increasing trend over time with stable variance outside of large spikes during quarters associated with COVID (Figure 1). Given the stable variance, a log transformation was not very necessary but did help address the variance spike during COVID when modeling.

Figure 1: Total direct maintenance cost per air hour over time

Limitations of this data include reporting corrections which are shown as negative spending on maintenance and negative hours flown as well as a large amount of unreported data on smaller craft. To address this, we removed the reporting corrections by filtering out hours flown less than or equal to 0 and dropped rows with unreported data in the hours flown and total direct maintenance cost fields. The resulting set contains only positive costs in order to enable a log transformation. Removing reporting corrections may artificially increase the predictions made and introduce additional bias, which should be noted when using the produced models.

3 Methods

A quarterly time series was created using frequency 4, since there are four quarters in each year. The original time series for each manufacturer was first plotted to examine trend, variability, and possible seasonal behavior. Because both maintenance cost series showed some variance over time, including large spikes during COVID, a logarithmic transformation to reduce the effect of these extreme values and help stabilize the variance was used before modeling by $X_t = log(X_t^*)$, where X_t^* is the original maintenance cost per air hour. In the transformed plots, the overall shape of the series remains the same, but the differences between large and small values are reduced (Figure 2). To account for quarterly seasonality, seasonal differencing was applied to the log-transformed series using lag 4: $\nabla_4 X_t = X_t - X_{t-4}$.

A lag-1 difference was then applied to remove the upward linear trend, resulting in two lags applied to the series: $\nabla_1 \nabla_4 X_t$. The first and seasonal differencings correspond to $d = 1$ and $D = 1$, respectively. After these differencings, the data appears to be stationary.

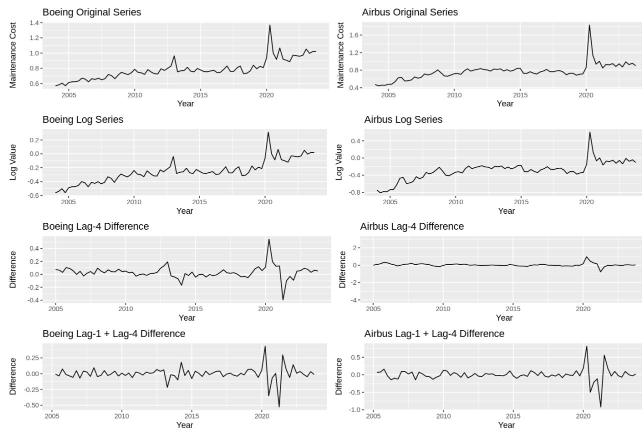

Figure 2: Effects of each transform on the data

3.1 SARIMA

The ACF and PACF plots of the differenced series were examined to guide model selection. The ACF plot shows a noticeable spike at lag 1, which suggests the presence of a moving average component. At the same time, the PACF plot does not show strong or consistent spikes, indicating that an autoregressive component is not necessary (Figure 3 and Figure 4). Based on these observations, a model with a non-seasonal MA(1) term is appropriate. Because the data is quarterly, a seasonal MA(1) component is also included.

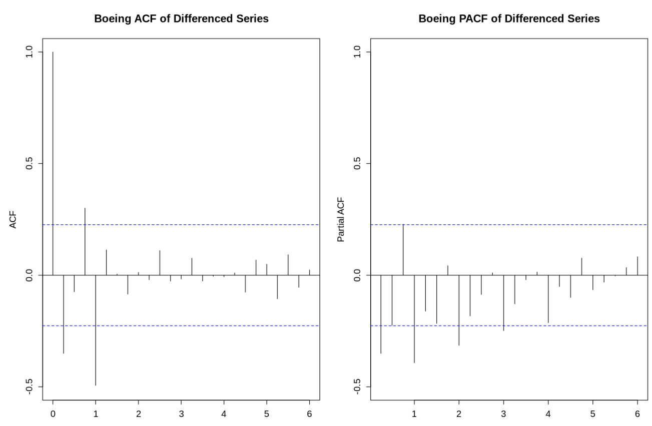

Figure 3: Boeing Differenced Series ACF and PACF plots

This leads to the selection of the $SARIMA(0,1,1)\times(0,1,1)_{s=4}$ model. This model is consistent with the patterns observed in the ACF and PACF plots and is also supported by information criteria such as AIC, BIC, and AICC. Several other candidate SARIMA models for each Boeing and Airbus were also compared using AIC, BIC, and AICC. These criterion confirmed the initial selected model for both Boeing and Airbus, $SARIMA(0,1,1)\times(0,1,1)_{s=4}$ was the best option.

Model coefficients were estimated separately for each Boeing and Airbus using the `arima( )` function in R. Parameter inference was conducted using estimated coefficients, standard errors, z-values, p-values, and 95% prediction intervals. Residual diagnostics were used to assess whether the fitted models were adequate. These residual checks included residual time plots, residual ACF

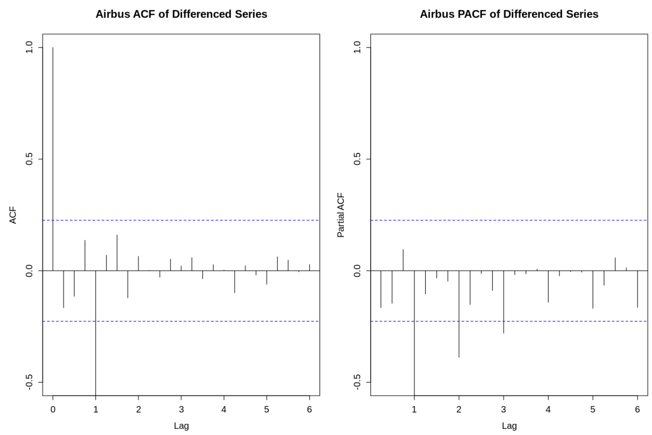

Figure 4: Airbus Differenced Series ACF and PACF plots

and PACF plots, histograms, Q-Q plots, Shapiro-Wilk normality tests and Ljung-Box tests. The goal was to determine whether the residuals behaved like white noise, meaning they should have approximately constant variance, no obvious time pattern, and no remaining autocorrelation. These tests indicated the residuals are uncorrelated, but not normally distributed. Better modeling or outlier handling may be needed (Table 1).

Table 1: P-Values of Residual Diagnostics for SARIMA Fits

Normality Test	Box-Ljung test $m = 8$	Box-Ljung test $m = 12$	Shapiro-Wilk
Boeing	0.3178	0.6483	6.81e-12
Airbus	0.2443	0.5651	1.59e-13
Implication	Uncorrelated	Uncorrelated	NOT Normally Distributed

The residual time plots for both manufacturers indicate that their respective $SARIMA(0,1,1)\times(0,1,1)_{s=4}$ model captures the main trend and seasonal structure, since the residuals fluctuate around zero with no clear systematic pattern (Figure 5). However, a large positive residual around 2020 suggests the presence of an outlier or unusual shock that was not captured by the model. This makes sense considering the impact COVID had on the aviation industry. Though residuals are mostly consistent with white noise, this spike suggests the 2020 observation may affect normality and prediction interval behavior. The residual ACF and PACF do not show meaningful spikes outside the significance limits, suggesting that the residuals are approximately uncorrelated and further support the assumption that they model white noise.

The residual histogram for Boeing indicates that most residuals are centered near zero, but the distribution is right-skewed as result of a large positive residual (Figure 5). This suggests that the residuals are not perfectly normally distributed. Airbus exhibits a similar right skew with a large outlier.

The Q-Q plot shows that most residuals follow the normal reference line reasonably well, but the extreme upper-tail point deviates strongly. This indicates that the normality assumption is weakened by the outlier likely from the COVID spike.

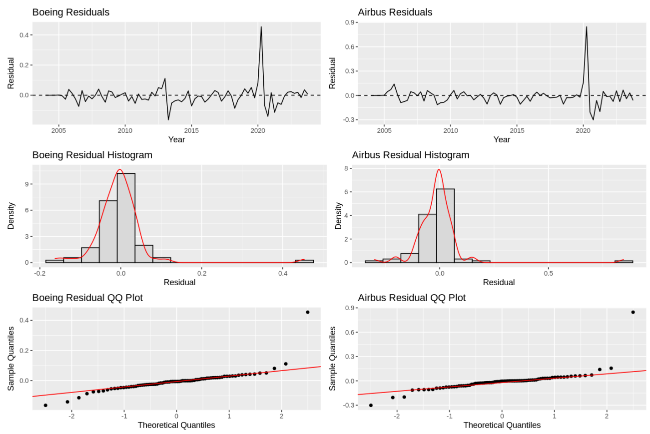

Figure 5: Residual plot and histogram of Boeing and Airbus fitted SARIMA models

Forecasts were generated for the next four quarters. The 95% prediction intervals were first calculated on the log scale and then back-transformed to the original scale (Figure 6). Forecast tables were produced for both Boeing and Airbus, including forecasted values and lower and upper 95% prediction interval limits.

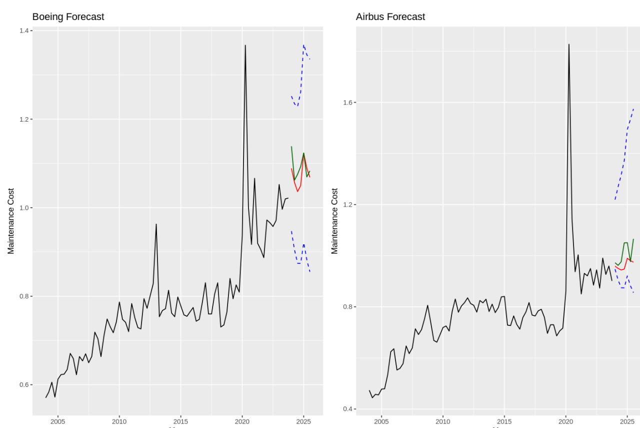

Figure 6: Forecast plot of Boeing and Airbus fitted SARIMA models with 95% prediction intervals and true data

The SARIMA forecasts for both Boeing and Airbus indicate a continuation of the upward trend in maintenance costs. Boeing shows a more stable pattern with relatively narrow 95% prediction prediction intervals, suggesting higher confidence in the forecasts. In contrast, Airbus exhibits greater variability and a larger spike around 2020, resulting in wider 95% prediction intervals and increased uncertainty. The results suggest that both manufacturers are expected to experience rising maintenance costs, though Airbus forecasts are less precise due to higher variability in the data and the larger impact of COVID related outliers.

3.2 ARIMAX

An Autoregressive Integrated Moving Average with exogenous inputs (ARIMAX) model expands upon the SARIMA model by including additional predictors which, along with time, may affect the response. A general ARIMAX formulation can be written as Equation (1).

$$X_t = \beta_0 + \sum_{i=1}^{p} \phi_i X_{t-i} + \sum_{j=1}^{q} \theta_j Z_{t-j} + \sum_{k=1}^{m} \beta_k P_{t-k} + \epsilon_t \tag{1}$$

This ARIMAX model expands upon the previously fit SARIMA model by including total cost of flight operations β^{ops}, which includes wages and fueling costs, amount of utilization β^{util}, and the number of carriers active for that quarter β^{ncar}. These variables are first constructed into a clean matrix of predictors which are each lag-1 differenced to match the series in the `make_xreg_base( )` function. Flagged anomalies were also included as exogenous variables. Anomalies were identified as residuals from the SARIMA model which are further than 2.5 standard deviations from the mean (Equation (2)). These identifications were made using the `zscore_anom` function. This function identified anomalies at 2020 Q_1 and Q_3 for Airbus and 2020 Q_1 for Boeing . Additional anomaly flags were placed to identify the rest of the 4 quarters of COVID: Q_1 and Q_2 flags were placed to identify the initial drop in flight demand, or 'shock' to the market, and Q_3 and Q_4 flags were placed to identify the later recovery seen in the increase in maintenance costs required to prepare previously grounded aircraft quickly for a return in demand. These include the anomalies flagged in the `zscore_anom` function (Equation 2).

$$A_t = \mathbf{1}\left(\left| \frac{e_t - \bar{e}}{s_e} \right| > 2.5 \right) \tag{2}$$

Initial data transformations for this fit mirror those used for the SARIMA: first, the data is log transformed, then differenced to remove trend and seasonality. The differencing may pose a limitation to the model as this effectively removes 5 data points from a predictor set of 80 data points. These lags are applied directly to the data series as $d = 1$ and $D = 1$ in the arima( ) calls for both the Boeing and Airbus models. Based on AICC and BIC, the selected ARIMAX models used $(1, 1, 1) \times (0, 1, 1)_{s=4}$ for Boeing and $(0, 1, 1) \times (0, 1, 1)_{s=4}$ for Airbus. The ARIMAX models outperformed their respective SARIMA models in AICC and BIC despite having more predictors (Table 5). The resulting models contain coefficients for these SARIMA values as well as for each of the included exogenous variables and the COVID indicator flags.

Airbus Model:

$$\begin{aligned}\nabla\nabla_4 \log(X_t) = 0.2221\, \nabla\nabla_4 \log(\beta^{\text{ops}}_{t-1}) \\ - 0.4720\, \nabla\nabla_4 \log(\beta^{\text{util}}_{t-1}) + 0.4838 \cdot \beta^{\text{shock}}_t \\ + Z_t - 0.6711 Z_{t-1} - Z_{t-4} + 0.6711 Z_{t-5}\end{aligned} \tag{3}$$

where $\nabla = (1 - B)$ is the first difference operator, $\nabla_4 = (1 - B^4)$ is the seasonal difference operator. Both recovery and $\hat{\theta}_1$ are removed from the model as they are not statistically significant with p-values > 0.05 (Table 6).

Boeing Model:

$$\nabla\nabla_4 \log(X_t) = -0.4203\, \nabla\nabla_4 \log(\beta_{t-1}^{\text{util}}) + 0.2782 \cdot \beta_t^{\text{shock}} + Z_t - 0.7243 Z_{t-1} - Z_{t-4} + 0.7243 Z_{t-5} \quad (4)$$

For Boeing, ϕ_1, log of operation costs, log of number of carriers, and recovery are not statistically significant (Table 6).

The response variable $log(X_t)$ is `log(MAINT_PER_AIR_HR)`, or the log of the dollar cost of maintenance per air hour. ARMA coefficients govern the autocorrelation structure of the log series after a seasonal lag ∇_4 and a linear trend lag ∇_1 have been applied. Coefficients are interpreted as multiplicative factors because the predicted value $log(X_t)$ is on the log scale. Continuous predictor coefficients for operation cost, utilization, and number of carriers are the percentage change in maintenance cost per air hour per 1% change in the respective predictor. Binary indicator coefficients for shock and recovery represent the approximate percentage acceleration or deceleration in cost growth during those quarters. For Airbus, the shock coefficient 0.2728 represents a $100\% - 27.28\% = 72.72\%$ faster growth in maintenance costs than seasonal changes would suggest (Box and Tiao 1975).

The seasonal MA(1) coefficient was estimated for both models as $\hat{\Theta}_1 = -1$, which lies on the boundary of the unit circle on the complex plane (Equation 3 and Equation 4). Since this value does not satisfy $|\Theta_1| < 1$, the fitted model is not invertible. This model may be over-differenced in the seasonal component, or need a different seasonal MA lag. The confidence interval for $\hat{\Theta}_1$ is wide, ranging from almost -2 to -0.14 for Airbus [1] which includes the invertability region and extends beyond.

Despite these limitations in the model, the residual series does appear to be white noise according to the SACF and SPACF plots. The residuals for Airbus are, however, not normally distributed as indicated by the histograms, QQ plots, and diagnostics scores (Figure 7). Boeing fares better and is consistent with the SARIMA conclusion that predictions for this series are more reliable as the QQ plot and rough normal distribution of the residuals suggest though the residuals from the Boeing ARIMAX fit still fail the Ljung-Box and Shapiro-Wilk normality tests. Large outliers of high residual values lead the residual series to not be IID (Table 2).

Table 2: P-Values of Residual Diagnostics for ARIMAX Fits

Normality Test	Box-Ljung test $m = 8$	Box-Ljung test $m = 12$	Shapiro-Wilk
Boeing	$< 2.2e-16$	0.3912	$5.657e-05$
Airbus	0.005084	0.0573	$2.109e-10$
Implication	Correlated	Uncorrelated	NOT Normally Distributed

The forecasts from these models contain the true data within their 95% prediction interval

[1] Boeing 95% $\hat{\Theta}_1$ Confidence interval: (-1.1864, -0.8135) and Airbus 95% $\hat{\Theta}_1$ Confidence interval: (-1.8525, -0.1475)

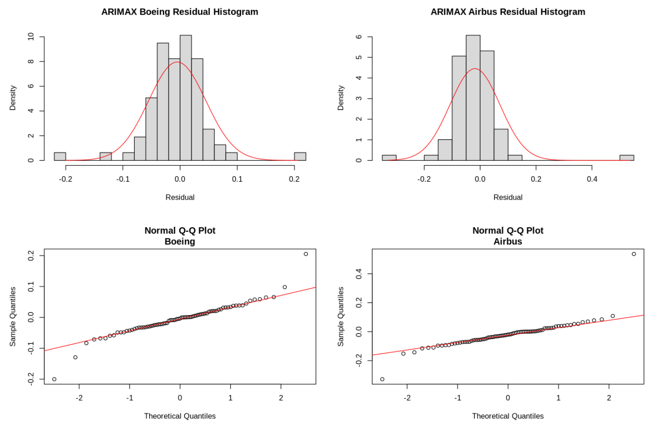

Figure 7: Histograms and QQ plots of Boeing and Airbus fitted ARIMAX models

though the model underestimates costs overall, though most notably for Boeing (Figure 8). Despite the predictions themselves seeming further from the true data compared to the SARIMA model, the 95% prediction intervals for both models is narrower while still including the true values, indicating a more confident fit.

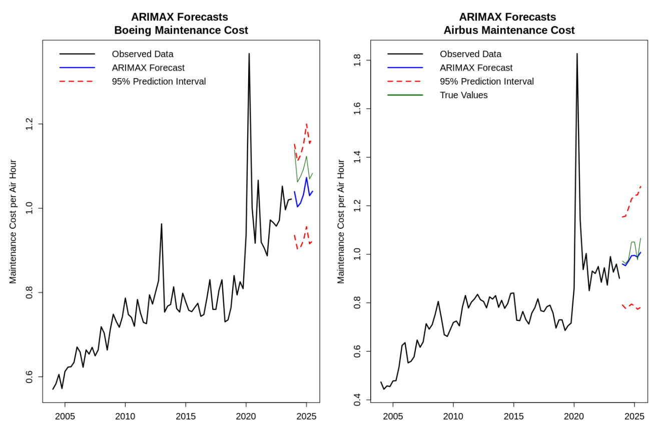

Figure 8: Forecast plot of Boeing and Airbus fitted ARIMAX models with 95% prediction intervals and True data

Further work could be done to better model the predictors such that they are better covariates for the series. This could be done by adding seasonal differencing or choosing better models than a naive linear fit. Additionally, the value of $\hat{\Theta}_1$ could be better predicted by addressing the high variance across seasons.

3.3 RNN with LSTM

A Recurrent Neural Network model (RNN) is a category of Neural Network models that specializes in processing sequential data such as text or Time Series by using past predictions from the Network to inform future forecasts. A common issue due to the reliance of past predictions in RNNs is the vanishing gradient problem. As the length of time steps between data points increases, RNN's will frequently "forget" important information if the information was gleaned from data too far in the past. To overcome this problem, an architecture called a Long Short-Term Memory (LSTM) was created. This structure has input, output, and forget gates to control the flow of weights from the previous data point until no longer useful.

This model used the same dataset as the SARIMA and ARIMAX models, but was expanded to include the synthetic variable: `Time_from_creation`, defined time between when a plane model was created and the present report averaged across the manufacturers' entire present portfolio of plane models. This effectively gives the average age of each manufacturers' aircraft. A key difference between how the data was used in RNN versus the SARIMA and ARIMAX models is that RNN's do not run properly on stationary data, so to account for this we did not take a Lag-1 or seasonal difference prior to create a stationary series prior to being fed into the model. The purpose of this is to allow the model to learn the inherent trends rather than potentially lose information when removing the trends then adding them back during forecasting. The data was first separated

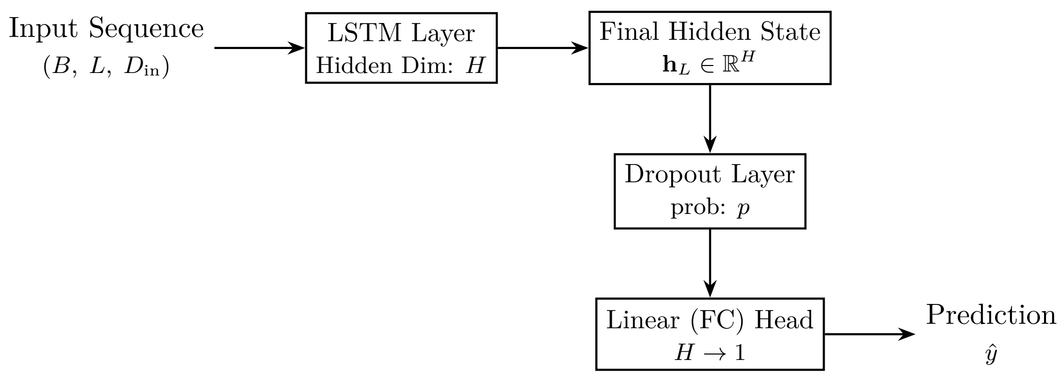

Figure 9: LSTM-RNN architecture used for maintenance cost forecasting. B = batch size, L = lookback window, D_{in} = number of input features, H = hidden dimension.

into numeric and categorical variables. The numerical variables are split along the train-validation index and scaled using the `StandardScaler( )` from the `sklearn` library in Python. These scaled features are then layered onto the entire dataset's numerical features to prevent data leakage. Next helper functions are created to construct the lookback sequences for the predictors and response variables as well as a function to create the train-val-test indexes in the dataset. Created using the `PyTorch` library, the model is structured such that it first forward propagates into a LSTM with a node count equal to the tuned hyperparameter labeled `hidden`, indicating the number of nodes in the hidden layer. Next, the LSTM outputs are dropped out at the hyperparameter rate `dropout`, before finally going through a fully connected layer also sized according to the `hidden` hyperparameter (Figure 9). After the model has completed a forward propagation pass, it is then back-propagated using an MSE loss function and the `Adam` algorithm as its optimizer with a learning rate `lr` similarly validated over.

After validating across 192 hyperparameter configurations, the final RNN model hyperparameters were selected to be: `lookback`: 16, `hidden`: 64, `lr`: 0.005, `dropout`: 0.2, `batch_size`: 16. After 100 epochs of training, this model produced the lowest test set $RMSE$ of 0.0316, and had a total parameter count of 72,833. Despite this, the final RNN model still does not have normally

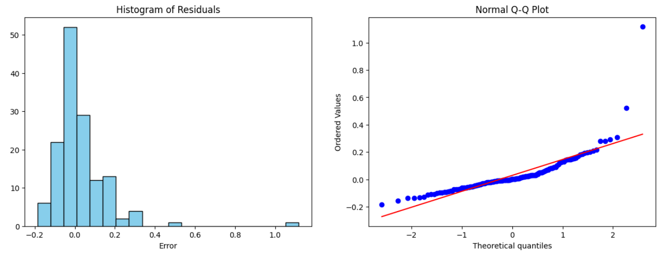

Figure 10: Aggregate Normality plots of Residuals From RNN Predicted Series

distributed residuals. A Shapiro-Wilk test returns a p-value of < 0.00001, and the QQ-Plot shows major divergence from normality at the graph's ends and a small but noticeable curve throughout (Figure 10).

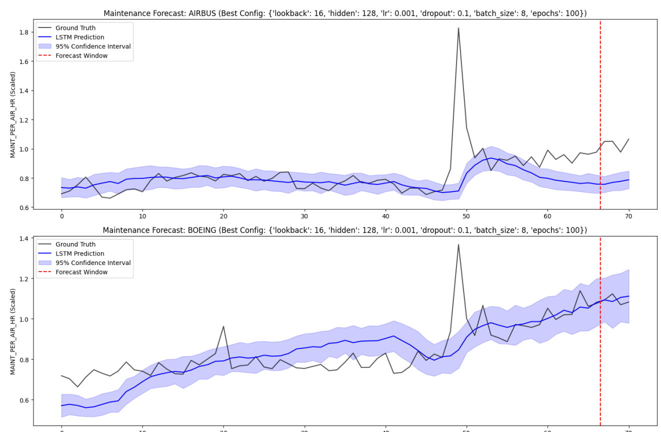

Figure 11: RNN Forecasts against Standard Scaled data for both Boeing and Airbus with 95% prediction bands

4 Results

All three models successfully captured the upward trend in maintenance costs for both manufacturers, though each differed in forecast accuracy and residual behavior. Forecast performance on the seven-quarter test set is summarized in Table 3 and visualized alongside true values in Figure 12.

For Boeing, model performance improved with complexity. The SARIMA baseline produced the largest errors, the ARIMAX reduced these, and the RNN achieved the best performance. The SARIMA and ARIMAX forecasts for Boeing both fall within their respective 95% prediction intervals, though both models underestimate costs in the most recent quarters (Figures 6 and 8).

For Airbus, the pattern was reversed for the RNN. Here, more complexity did not improve model performance. ARIMAX outperformed SARIMA substantially and the RNN performed worst of the three, suggesting the model may have overfit to Boeing's patterns during training. A single RNN was trained using all data even though the data given to the RNN included which manufacturer the values corresponded to. The ARIMAX model is therefore the recommended approach for Airbus forecasting given its strong predictive accuracy and the added benefit of coefficient interpretability.

Table 3: Test set forecast performance by model and manufacturer

Manufacturer	Model	RMSE	MAE
Boeing	SARIMA	0.0794	0.0656
	ARIMAX	0.0620	0.0592
	RNN	**0.0362**	**0.0244**
Airbus	SARIMA	0.0585	0.0452
	ARIMAX	**0.0378**	**0.0297**
	RNN	0.0847	0.0781

Note: Bold entries indicate the best-performing model for each manufacturer. RMSE and MAE are computed on the log-scale test set of $n = 7$ quarters. Lower values indicate better forecast accuracy.

Residual diagnostics were consistent across all three model classes: the Ljung-Box tests indicate uncorrelated residuals in most cases, but Shapiro-Wilk tests reject normality for every fitted model (Tables 1 and 2). This non-normality is driven primarily by the large outliers during the COVID quarters of 2020 which no model was completely able to handle, as seen in each of their residual plots. As a result, the stated 95% prediction intervals should be interpreted with caution, as they rely on the normality assumption which these models do not satisfy.

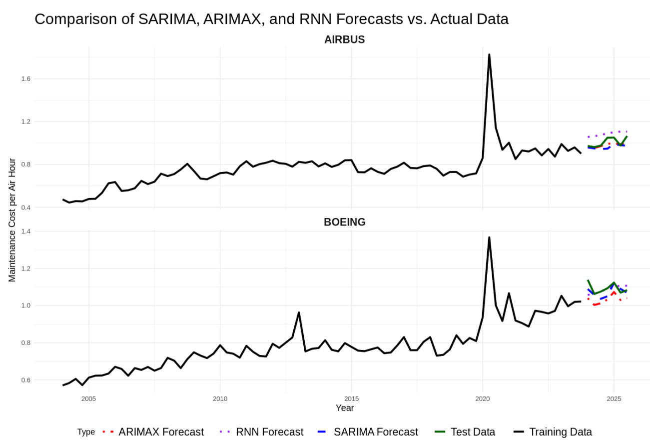

Figure 12: Forecast plot of all Fitted Models (SARIMA, ARIMAX, RNN) on both Boeing and Airbus with True data

5 Conclusions and Discussion

This was a very interesting chance for us to compare the abilities of traditional statistical models: SARIMA and ARIMAX against a deep learning model like an RNN, especially in the use case of such volatile data as aviation financial forecasting. Our most interesting finding when comparing these models was the difference in forecasting abilities between manufacturers. As can be seen in the table below, when forecasting Boeing the RNN model outperformed either the SARIMA or the ARIMAX, and is actually the best performing of all 6 models with an $RMSE$ 2.15x lower than the next best model. However, when forecasting Airbus, it was the worst performing out of all 6 models created for this study, showing the continuing need for models beyond the scope of Deep Learning even in the modern age.

By diving into more difficult, chaotic data using both statistical and deep learning models and getting a chance to compare their abilities, we believe we improved on our skillset to analyze Time Series data in future applications. While the RNN model did outperform the other models when forecasting Boeing, the explainability and simplicity of the SARIMA and ARIMAX still allowed for strong forecasting accuracy. In fact, when averaged across both manufacturers our best performing model was the ARIMAX model with an average $RMSE$ of 0.0499 against the RNN and SARIMA industry wide $RMSE$ of .0561 and .06895, respectively. This could suggest that with more time and more data, we could find interesting differences in our forecasting approach between individual companies vs the wider aviation industry.

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References

Box, G. E. P. and Tiao, G. C. (1975). Intervention analysis with applications to economic and environmental problems. *Journal of the American Statistical Association*, 70(349):70–79.

Bureau of Transportation Statistics (2025a). Air carrier financial : Schedule p-5.2. https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FMK. Accessed: 2026-04-28.

Bureau of Transportation Statistics (2025b). Aircrafttypes. https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=GDD. Accessed: 2026-04-28.

Appendix

A.1 Analysis details and model diagnostics

Table 4: SARIMA$(0, 1, 1) \times (0, 1, 1)_4$ coefficient estimates for Boeing and Airbus

Parameter	Boeing	Airbus
$\hat{\theta}_1$	-0.5079	-0.3237
$\hat{\Theta}_1$	-1.0000	-0.9335

Note: The exact estimate of $\hat{\Theta}_1$ is -0.9999972.

Table 5: Information criteria for candidate Boeing and Airbus models

Manufacturer	Model	AIC	BIC	AICC
Boeing	**ARIMAX**$(1, 1, 1) \times (0, 1, 1)_4$	-196.78	-176.04	-193.97
	ARIMAX$(0, 1, 1) \times (0, 1, 1)_4$	-196.00	-177.57	-193.78
	SARIMA$(0, 1, 1) \times (0, 1, 1)_4$	-166.45	-159.54	-166.11
Airbus	**ARIMAX**$(0, 1, 1) \times (0, 1, 1)_4$	-111.18	-92.74	-108.96
	SARIMA$(0, 1, 1) \times (0, 1, 1)_4$	-86.50	-79.59	-86.16
	ARIMAX$(0, 1, 1) \times (0, 1, 0)_4$	-76.14	-60.01	-74.44

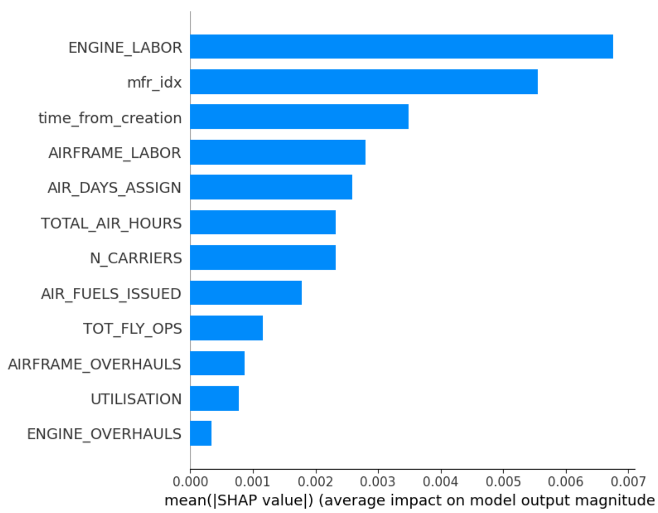

Figure 13: RNN Variable Importance: SHAP Values

Table 6: ARIMAX coefficient estimates for Boeing $(1,1,1) \times (0,1,1)_4$ and Airbus $(0,1,1) \times (0,1,1)_4$

	Parameter	Estimate	SE	z	p-value	95% CI
Boeing	$\hat{\phi}_1$	-0.2600	0.1462	-1.779	0.0753	$(-0.5465,\ 0.0265)$
	$\hat{\theta}_1$	-0.7243	0.1119	-6.475	< 0.001	$(-0.9436,\ -0.5051)$
	$\hat{\Theta}_1$	-1.0000	0.0951	-10.513	< 0.001	$(-1.1864,\ -0.8135)$
	$\hat{\beta}_{ops}$	0.0678	0.0448	1.514	0.1299	$(-0.0200,\ 0.1556)$
	$\hat{\beta}_{util}$	-0.4203	0.0982	-4.281	< 0.001	$(-0.6127,\ -0.2279)$
	$\hat{\beta}_{ncar}$	0.1496	0.0982	1.524	0.1274	$(-0.0428,\ 0.3420)$
	$\hat{\beta}_{shock}$	0.2782	0.0351	7.927	< 0.001	$(0.2094,\ 0.3470)$
	$\hat{\beta}_{rec}$	-0.0998	0.0634	-1.574	0.1155	$(-0.2241,\ 0.0245)$
Airbus	$\hat{\theta}_1$	-0.6711	0.0869	-7.724	< 0.001	$(-0.8414,\ -0.5008)$
	$\hat{\Theta}_1$	-1.0000	0.4349	-2.299	0.0215	$(-1.8525,\ -0.1475)$
	$\hat{\beta}_{ops}$	0.2221	0.0904	2.456	0.0140	$(0.0449,\ 0.3994)$
	$\hat{\beta}_{util}$	-0.4720	0.1457	-3.240	0.0012	$(-0.7575,\ -0.1865)$
	$\hat{\beta}_{ncar}$	-0.0973	0.2084	-0.467	0.6405	$(-0.5058,\ 0.3112)$
	$\hat{\beta}_{shock}$	0.4838	0.0722	6.700	< 0.001	$(0.3423,\ 0.6253)$
	$\hat{\beta}_{rec}$	-0.0524	0.1164	-0.450	0.6528	$(-0.2805,\ 0.1757)$

Note: Parameters not significant at the $\alpha = 0.05$ level ($\hat{\phi}_1$, $\hat{\beta}_{ops}$, $\hat{\beta}_{ncar}$, $\hat{\beta}_{rec}$ for Boeing; $\hat{\beta}_{ncar}$, $\hat{\beta}_{rec}$ for Airbus) were removed from the final models.

A.2 Key R scripts for time series analysis

This section provides the main R scripts used for time series analysis in all models. Only the essential parts relevant to time series modeling and analysis are included.

```
# Load necessary libraries
library(ggplot2)
library(tidyverse)
library(dplyr)
library(gridExtra)
library(patchwork)
library(car)

# Load dataset from hoseted github
file_url <- "https://github.com/.../T_F41SCHEDULE_P52.csv"
craft_lookup_url <- "https://github.com/.../craft_codes_lookup/T_AIRCRAFT_TYPES.csv"

# data Cleaning
df <- left_join(df_raw, craft_type_lookup, by = c("AIRCRAFT_TYPE" = "AC_TYPEID")) %>%
  filter(YEAR > 2003) %>%
  filter(MANUFACTURER == "AIRBUS INDUSTRIE" | MANUFACTURER == "BOEING") %>%
  filter(AIRCRAFT_CONFIG %in% c(1, 2))%>%
  select(# Identifiers
    YEAR, QUARTER, MANUFACTURER,
    UNIQUE_CARRIER, CARRIER_NAME, CARRIER_GROUP,
```

```
    AIRCRAFT_TYPE, SHORT_NAME, LONG_NAME, AC_GROUP,
    AIRCRAFT_CONFIG, REGION, BEGIN_DATE,
    TOTAL_AIR_HOURS, AIR_DAYS_ASSIGN, AIR_FUELS_ISSUED, TOT_FLY_OPS,
    TOT_DIR_MAINT, TOT_FLT_MAINT_MEMO, TOT_AIR_OP_EXPENSES, AP_MT_BURDEN,
    AIRFRAME_LABOR, ENGINE_LABOR,
    AIRFRAME_REPAIR, ENGINE_REPAIRS,
    AIRFRAME_MATERIALS, ENGINE_MATERIALS,
    AIRFRAME_OVERHAULS, ENGINE_OVERHAULS) %>%
    mutate(TIME_POINT = as.numeric(YEAR + 0.25*(QUARTER-1))) %>%
    filter(!is.na(TOT_DIR_MAINT)) %>%
    filter(!is.na(TOTAL_AIR_HOURS)) %>%
    filter(!is.na(TOT_FLY_OPS)) %>%
    filter(TOTAL_AIR_HOURS > 0) %>%
  group_by(MANUFACTURER, YEAR, QUARTER, TIME_POINT) %>%
  summarise(
    TOT_DIR_MAINT    = sum(TOT_DIR_MAINT,    na.rm = TRUE),
    TOTAL_AIR_HOURS  = sum(TOTAL_AIR_HOURS,  na.rm = TRUE),
    TOT_FLY_OPS      = sum(TOT_FLY_OPS,      na.rm = TRUE),
    AIR_DAYS_ASSIGN  = sum(AIR_DAYS_ASSIGN,  na.rm = TRUE),
    AIR_FUELS_ISSUED = sum(AIR_FUELS_ISSUED, na.rm = TRUE),
    AIRFRAME_LABOR   = sum(AIRFRAME_LABOR,   na.rm = TRUE),
    ENGINE_LABOR     = sum(ENGINE_LABOR,     na.rm = TRUE),
    AIRFRAME_OVERHAULS = sum(AIRFRAME_OVERHAULS, na.rm = TRUE),
    ENGINE_OVERHAULS   = sum(ENGINE_OVERHAULS,   na.rm = TRUE),
    N_CARRIERS       = n_distinct(UNIQUE_CARRIER),
    .groups = "drop"
  ) %>%
  mutate(
    MAINT_PER_AIR_HR = TOT_DIR_MAINT / (TOTAL_AIR_HOURS * 1000),
    UTILISATION      = if_else(AIR_DAYS_ASSIGN > 0,
                               (TOTAL_AIR_HOURS * 1000) / AIR_DAYS_ASSIGN, NA_real_)
  ) %>%
  arrange(MANUFACTURER, TIME_POINT)

# Convert to time series object
df_boeing <- df_sarima %>%
  filter(MANUFACTURER == "BOEING") %>%
  arrange(TIME_POINT)

df_airbus <- df_sarima %>%
  filter(MANUFACTURER == "AIRBUS INDUSTRIE") %>%
  arrange(TIME_POINT)

y.t <- ts(df_boeing$MAINT_PER_AIR_HR,
          start = c(floor(min(df_boeing$TIME_POINT)), 1),
          frequency = 4)

y.t_airbus <- ts(df_airbus$MAINT_PER_AIR_HR,
                 start = c(floor(min(df_airbus$TIME_POINT)), 1),
```

```
                frequency = 4)
```

SARIMA core code

```
# Plot the time series NOTE: The following is only for Boeing but the process is the same for Airbus
plot.ts(y.t,
        main = "Boeing Maintenance Cost per Air Hour (Train)",
        xlab = "Year",
        ylab = "Maintenance Cost per Air Hour")

# Plot ACF and PACF
par(mfrow = c(1,2), mex = 0.75)
acf(ly.t_d1s4,
     ylim = c(-0.5, 1),
    lag.max = 24,
    main = "Boeing ACF of Differenced Series")
pacf(ly.t_d1s4,
     ylim = c(-0.5, 1),
     lag.max = 24,
     main = "Boeing PACF of Differenced Series")
par(mfrow = c(1,1))

# Fit an ARIMA model
fit_011_011 <- arima(ly.t,
                     order = c(0,1,1),
                     seasonal = list(order = c(0,1,1), period = 4),
                     include.mean = FALSE)

# Check residuals
fit_bst <- fit_011_011
w.t <- fit_bst$resid

# Forecast next 12 values
prd <- predict(fit_bst, n.ahead = length(x.t_test))

log_fc_boeing_sarima <- as.numeric(prd$pred)
log_se_boeing_sarima <- as.numeric(prd$se)


fc_boeing_sarima <- exp(log_fc_boeing_sarima)

lower_95 <- pmax(exp(log_fc_boeing_sarima - 1.96 * log_se_boeing_sarima), min(y.t))
upper_95 <- pmin(exp(log_fc_boeing_sarima + 1.96 * log_se_boeing_sarima), max(y.t) * 2)

fc_start <- tsp(y.t)[2] + 1/4

fc.ts <- ts(fc_boeing_sarima,
            start = fc_start,
            frequency = 4)
```

```
lower.ts <- ts(lower_95,
               start = fc_start,
               frequency = 4)

upper.ts <- ts(upper_95,
               start = fc_start,
               frequency = 4)

# Plot forecast
plot.ts(y.t,
        xlim = c(start(y.t)[1], end(y.t)[1] + 2),
        main = "SARIMA Forecasts for Boeing Maintenance Cost",
        xlab = "Year",
        ylab = "Maintenance Cost per Air Hour",
        col = "black",
        lwd = 2)
```

ARIMAX and Anomaly detection core code

```
# NOTE: The following code only models Boeing. Airbus modeling follows the same procedure
tp_b <- df_sarima_b$TIME_POINT
covid_shock_b    <- as.integer(tp_b >= 2020.00 & tp_b <= 2020.25)
covid_recovery_b <- as.integer(tp_b >= 2020.50 & tp_b <= 2020.75)

# log predictors to put on same scale as series
df_sarima_b <- df_sarima_b %>%
  mutate(log_ops  = log(TOT_FLY_OPS),
         log_util = log(UTILISATION),
         log_ncar = log(N_CARRIERS))

pred_cols <- c("log_ops", "log_util", "log_ncar")
lag_spec  <- c(log_ops = 1L, log_util = 1L, log_ncar = 1L)

# Create Predictor matrix
xreg_b_train <- cbind(make_xreg_base(df_sarima_b, pred_cols, lag_spec),
                shock = covid_shock_b,
                recovery = covid_recovery_b)

# Fit ARIMAX model using the predictor matrix with anomaly flags
fit_arimax_b <- arima(y_b_fit, order = c(0, 1, 1),
                      seasonal = list(order = c(0, 1, 1), period = 4),
                      xreg = xreg_b_fit, include.mean = FALSE)

# validate normality: Residuals Plot
par(mfrow = c(1, 2), mex = 0.75)
plot.ts(w.t_b,
        main = "Residuals from Boeing \nARIMAX Best Fit",
        xlab = "Year",
        ylab = "Residual")
abline(h = 0, col = "grey", lty = "dashed")
```

```
# Residuals ACF PACF
par(mfrow = c(1,2), mex = 0.75)
acf(residuals(fit_arimax_b), ylim = c(-0.5, 1), lag.max = 24,
    main = "Boeing Residuals ACF")
pacf(residuals(fit_arimax_b), ylim = c(-0.5, 1), lag.max = 24,
     main = "Boeing Residuals PACF")
par(mfrow = c(1,1))

# validate normality: Normality tests
n_par_b <- length(coef(boeing_best_fit))
Box.test(w.t_b, lag = 8,  type = "Ljung-Box", fitdf = n_par_b)
Box.test(w.t_b, lag = 12, type = "Ljung-Box", fitdf = n_par_b)
shapiro.test(w.t_b)

# Forecast
# Construct newxreg for Boeing: Matrix of predictors going forward
fit_boeing_ops <- lm(log_ops~TIME_POINT, data = df_sarima_b)
fit_boeing_util <-lm(log_util~TIME_POINT, data = df_sarima_b)
fit_boeing_ncar <-lm(log_ncar~TIME_POINT, data = df_sarima_b)

newxreg_boeing <- matrix(
  c(predict(fit_boeing_ops, pred_times),
    predict(fit_boeing_util, pred_times),
    predict(fit_boeing_ncar, pred_times),
    rep(0, n_ahead_forecast), # for shock
    rep(0, n_ahead_forecast)), # for recovery (All these are 0 because we don't expeck a shock)
  nrow = n_ahead_forecast, ncol = 5, byrow = FALSE
)

prd_boeing <- predict(boeing_best_fit, n.ahead = n_ahead_forecast, newxreg = newxreg_boeing)

log_fc_boeing <- prd_boeing$pred # These are on log scale
log_se_boeing <- prd_boeing$se   # These are on log scale

# Point forecast (mean, bias-corrected for log-normal distribution)
fc_boeing <- exp(log_fc_boeing + 0.5 * log_se_boeing^2)

# Prediction intervals on original scale
lower_boeing <- exp(log_fc_boeing - 1.96 * log_se_boeing)
upper_boeing <- exp(log_fc_boeing + 1.96 * log_se_boeing)

# Plot forecast
plot.ts(y_b,
        xlim = c(start(y_b)[1], end(y_b)[1] + n_ahead_forecast/4 + 0.5),
        main = "ARIMAX Forecasts \nBoeing Maintenance Cost",
        xlab = "Year",
        ylab = "Maintenance Cost per Air Hour",
        col = "black",
```

```
       lwd = 2)

lines(fc_boeing_ts, col = "blue", lwd = 2)
lines(lower_boeing_ts, col = "red", lty = 2, lwd = 2)
lines(upper_boeing_ts, col = "red", lty = 2, lwd = 2)
lines(x.t_test, col = "dark green")
```

Specialized Functions Created for ARIMAX

```
zscore_anom <- function(residuals, threshold = 2.5) {
  sigma_hat  <- sd(residuals)
  z_scores   <- residuals / sigma_hat
  time_axis  <- as.numeric(time(residuals))

  flag_idx   <- which(abs(z_scores) > threshold)
  flag_times <- time_axis[flag_idx]
  flag_z     <- as.numeric(z_scores)[flag_idx]

  data.frame(
    idx       = flag_idx,
    Quarter   = format(as.yearqtr(flag_times)),
    z_score   = round(flag_z, 3),
    direction = ifelse(flag_z > 0, "ABOVE trend", "BELOW trend")
  )
}

make_xreg_base <- function(df, cols, lags = setNames(rep(0L, length(cols)), cols)) {
  out <- matrix(NA_real_, nrow = nrow(df), ncol = length(cols),
                dimnames = list(NULL, cols))
  for (col in cols) {
    k <- lags[[col]]
    x <- as.numeric(df[[col]])
    if (k == 0L) out[, col] <- x
    else         out[, col] <- c(rep(NA_real_, k), x[seq_len(length(x) - k)])
  }
  out
}

build_anom_dummy <- function(flag_df, n, after = 1L) {
  dummy <- integer(n)
  for (i in flag_df$idx + 1L) {
```

```
    dummy[i : min(n, i + after)] <- 1L
  }
  dummy
}
```

RNN Core Code

```
# Model class initialization + structure
class MfrLSTM(nn.Module):
def __init__(self, input_dim, hidden_dim, dropout):
    super().__init__()
          self.lstm = nn.LSTM(input_dim, hidden_dim, batch_first=True)
    self.dropout = nn.Dropout(dropout)
    self.fc = nn.Linear(hidden_dim, 1)


      def forward(self, x):
          _, (hn, _) = self.lstm(x)
    out = self.dropout(hn[-1])
    return self.fc(out)

# Model Training Function
def train_robust_model(config, config_id):
    (X_train, y_train), (X_val, y_val), (X_test, y_test) =
      get_split_sequences(data_scaled,config['lookback'])

    train_loader = DataLoader(list(zip(X_train, y_train)),
      batch_size=config['batch_size'], shuffle=False)
   model = MfrLSTM(len(features), config['hidden'],
     config['dropout']).to(device)
   optimizer = optim.Adam(model.parameters(), lr=config['lr'])
   criterion = nn.MSELoss()


   best_val_loss = float('inf')


   # Create a unique filename for this specific hyperparameter set
   temp_filename = f'model_cfg_{config_id}.pth'
```

```
    for epoch in range(config['epochs']):
        model.train()
        for bx, by in train_loader:
            bx, by = bx.to(device).float(),
              by.to(device).float().unsqueeze(1)
            optimizer.zero_grad()
            outputs = model(bx)
            loss = criterion(outputs, by)
            loss.backward()
            optimizer.step()


        current_val_loss = evaluate_model(model, X_val, y_val, criterion)
        if current_val_loss < best_val_loss:
            best_val_loss = current_val_loss
            torch.save(model.state_dict(), temp_filename)


    # Load the best state FOR THIS CONFIG to run the robustness test
    model.load_state_dict(torch.load(temp_filename))
    test_robustness_loss = evaluate_model(model, X_test, y_test, criterion)


    return best_val_loss, test_robustness_loss, temp_filename


# Model Evaluation Function
def evaluate_model(model, X_data, y_data, criterion):
    model.eval()
    with torch.no_grad():
        # Ensure input is 3D: [Number of Samples, Lookback, Features]
        inputs = torch.tensor(X_data).to(device).float()


        # If X_data was [Lookback, Features] (1 sample), we must add the
        batch dimension
        if inputs.dim() == 2:
            inputs = inputs.unsqueeze(0)
```

```
        targets = torch.tensor(y_data).to(device).float().unsqueeze(1)


        outputs = model(inputs)
        loss = criterion(outputs, targets)


        return loss.item()


# Helper Function to create separate sequences for Airbus and Boeing
def create_mfr_sequences(df, lookback):
    X, y = [], []
    # Process each manufacturer separately to maintain sequence integrity
    for mfr in [0, 1]:
        mfr_data = df[df['mfr_idx'] == mfr].sort_values('TIME_POINT')
        vals = mfr_data[features + [target]].values


        if len(vals) > lookback:
            for i in range(len(vals) - lookback):
                X.append(vals[i : i + lookback, :-1])
                y.append(vals[i + lookback, -1])


    return np.array(X), np.array(y)


# Code to output QQ-Plot and Shapiro-Wilk Test
# QQ-Plot
stats.probplot(all_residuals, dist="norm", plot=ax[1])
ax[1].set_title("Normal Q-Q Plot")
plt.show()


# Shapiro-Wilk Test
sw_stat, sw_p = stats.shapiro(all_residuals)
print(f"\n--- Statistical Diagnostics ---")
print(f"Shapiro-Wilk Test: W={sw_stat:.4f}, p-value={sw_p:.4f}")
if sw_p > 0.05:
```

```
    print("Result: Residuals appear Normally Distributed (Fail to reject H0)")
else:
    print("Result: Residuals are NOT Normally Distributed (Reject H0)")
```